2311018

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” ” Word Embedding

1. Bag-of-Words Word2Vec/GloVe BERT
2.
3. Logistic Regression
4. Word2Vec GloVe
5. BERT Fine-tuning
6. Accuracy Macro-F1 Macro-F1
7.

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**2.1**

| / | | |
|---|---|---|
| | Windows 10/11 (x64) | |
| CPU | Intel Core i5, 4 8 | CPU |
| | 16 GB | |
| Python | 3.12.7 (Anaconda) | |
| PyTorch | 2.13.0+cpu | CUDA: False |
| Transformers | 5.x | BERT |
| scikit-learn | 1.9.0 | |
| gensim | 4.3.x | Word2Vec |
| NLTK | 3.9.x | |

**2.2**

**NYT** - `text` - `label` sports/politics/business

**AG News** text Word2Vec

**2.3**

- 80% 9,215 - 20% 2,304 - 42 - stratified split

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**3.1**

NYT - 42 - 20% -

**3.2**

1.
2. `\b\w+\b` tokenize
3. sklearn train_test_split

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**4.1 Accuracy**

**4.2 Macro-F1**

Macro-F1 F1 Score F1 Macro-F1

**Macro-F1 Accuracy**

sports 75% Accuracy business politics sports Accuracy 75% Macro-F1

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## Task 1 Bag-of-Words 30

**5.1**

Bag-of-Words " "

**5.2**

**5.2.1 Binary Bag of Words** Vocabulary |V| |V| - x = 1 - x = 0

**5.2.2 Word Frequency** Binary BoW Word Frequency - x = c c i

**5.2.3** BoW **Logistic Regression** - max_iter = 1000 - random_state = 42

**5.3**

| | Accuracy | Macro-F1 |
|---|---|---|
| Binary Bag of Words | 0.9848 | 0.9634 |
| Word Frequency | **0.9870** | **0.9687** |
| Count Vector | 0.9861 | 0.9677 |

61,021

Word Frequency BoW 61,021

**5.4**

1. NYT 75% BoW ”score” “game” ”government” “election”
2. BoW 61,021
3.

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## Task 2 Word Embedding 50

**6.1**

Word Embedding 100-dimensional word vectors 100

n e(w ), e(w ), ..., e(w )

**e(d) = (1/n) × Σ e(w )**

**6.2 Pre-trained GloVe glove.6B.100d**

GloVe Embedding glove.6B 100-dimensional GloVe word vectors

- http://nlp.stanford.edu/data/glove.6B.zip
- 400,000
- 100

**6.3 Word2Vec Trained on AG News**

AG News 100-dimensional Word2Vec Model - Gensim Word2Vec - 100 - 5 - 1 - 10 - 90,000 AG News

### 6.4 Word2Vec Trained on NYT

NYT 100-dimensional Word2Vec Model - 6.3 - 9,215 NYT

### 6.5

| | Accuracy | Macro-F1 |
|---|---|---|
| GloVe (Pre-trained) | **0.9813** | **0.9573** |
| Word2Vec (AG News) | 0.9774 | 0.9492 |
| Word2Vec (NYT) | 0.9783 | 0.9484 |

### 6.6

1. **GloVe** 40
2. NYT Word2Vec AG News
3. ” ” —— GloVe AG News
4.

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## Task 3 BERT 20

### 7.1

Fine-tuning - google-bert/bert-base-uncased - Tokenization max_length = 64 - 3 epochs

### 7.2

1. **padding** batch batch
2. **tokenize** BERT
3. epoch Macro-F1 checkpoint

### 7.3

| Epoch | Train Loss | Val Macro-F1 |
|---|---|---|
| 1 | 0.2786 | 0.9403 |
| 2 | 0.0457 | 0.9403 |
| 3 | 0.0152 | 0.9417 |

**7.4**

| | Accuracy | Macro-F1 |
|---|---|---|
| BERT-base-uncased max_len=64, 3 epochs | 0.9766 | 0.9469 |

**7.5**

1. **BERT** BERT Accuracy 0.9766 / Macro-F1 0.9469 Task 1 0.9870 / 0.9687
2. **max_length=64** BERT 49 8.0% 604
3. + 9,215 BERT-base 1.1 0.2786 0.0152 2 epoch
4. **BERT** " " BERT 49 Macro-F1 0.9469 160

**7.6**

"BERT " + Logistic Regression K

| K | Test Accuracy | Test Macro-F1 | |
|---|---|---|---|
| 10 | 0.8993 | 0.8147 | |
| 20 | 0.9314 | 0.8781 | |
| 40 | 0.9523 | 0.9044 | |
| 49 | 0.9557 | 0.9114 | BERT |
| 80 | 0.9696 | 0.9367 | |
| 160 | 0.9748 | 0.9491 | |
| 320 | 0.9800 | 0.9594 | |
| 604 | 0.9896 | 0.9771 | Task 1 |
| BERT | 0.9783 | 0.9509 | 49 |

1. 2. K = 49 BERT Macro-F1 0.9114 BERT 0.9509 4 3. BERT

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**8.1**

| | | | Accuracy | Macro-F1 |
|---|---|---|---|---|
| Task 1 | Binary BoW | 57,453 | 0.9848 | 0.9634 |
| Task 1 | Word Frequency | 57,453 | **0.9870** | **0.9687** |

| | | | Accuracy | Macro-F1 |
|---|---|---|---|---|
| Task 2 | GloVe (Pre-trained) | 100 | 0.9813 | 0.9573 |
| Task 2 | Word2Vec (AG News) | 100 | 0.9774 | 0.9492 |
| Task 2 | Word2Vec (NYT) | 100 | 0.9783 | 0.9484 |
| Task 3 | BERT (Fine-tuned) | 768 | 0.9766 | 0.9469 |

**8.2**

1. Word Frequency Macro-F1 0.9687 BERT 0.9469 2 3
2. **Macro-F1 Accuracy** BERT 5 Accuracy 0.98 Macro-F1
3. **business/politics** sports F1 0.99 business politics 0.93~0.97 ” ”
4. “ ” ” ”
5. BoW 24 70 BERT 105 2

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**9.1**

1.
2.
3. **BERT**

**9.2 ” ”**

Logistic Regression 110M BERT 49 Macro-F1 BERT 4 BERT ” ” ” ”

**9.3**

NYT 604 BERT max_length=64 49 8.0% ” ”

**9.4**

CPU BERT batch size “ ” ” ”

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### 10.1

```
HW-1/
   homework1.py              # Task 1:
   homework1_word2vec.py     # Task 2.2 & 2.3: Word2Vec
   homework1_glove.py        # Task 2.1: GloVe
   homework1_bert.py         # Task 3: BERT
   nyt.csv                   # NYT
   ag.csv                    # AG News
   task1_results.csv         # Task 1
   task2_results.csv         # Task 2 Word2Vec
   task2_glove_results.csv   # Task 2 GloVe
   task3_results.csv         # Task 3
```

### 10.2

```
pip install numpy pandas scikit-learn gensim transformers torch accelerate
```

**Task 1**

```
python homework1.py
```

**Task 2.1 GloVe glove.6B.100d.txt**

```
#  https://nlp.stanford.edu/data/glove.6B.zip     glove.6B/
python homework1_glove.py
```

**Task 2.2 & 2.3 Word2Vec**

```
python homework1_word2vec.py
```

**Task 3 BERT bert-base-uncased**

```
#  https://huggingface.co/google-bert/bert-base-uncased     bert-base-uncased/
python homework1_bert.py
```

### 10.3

- 42
- BERT 440MB HuggingFace
- GloVe 400MB Stanford